ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Date: 01/08/2026

Experiment 7: Logistic Regression (LR) Algorithm

Aim

To implement the Logistic Regression (LR) algorithm in Python for binary classification using an appropriate dataset.

Procedure

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Normalize the input features.
5. Create the Logistic Regression model.
6. Train the model using the training data.
7. Predict the class labels for the test data.
8. Evaluate the model using accuracy and confusion matrix.
9. Display the prediction results.

Algorithm

1. Start.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Scale the input features.
5. Initialize the Logistic Regression model.
6. Train the model using the training data.
7. Predict the class labels for the test data.
8. Display the predicted results and accuracy.
9. Stop.

Code

```python
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_breast_cancer()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Labels:")
print(y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Output

```
Predicted Labels:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 1 1 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 0 0]
Accuracy: 0.9736842105263158
Confusion Matrix:
[[41  2]
 [ 1 70]]
```

Result

The Logistic Regression (LR) algorithm was successfully implemented in Python. The model was trained and tested using the Breast Cancer dataset and achieved high classification accuracy for binary classification.